Module 2 Report Addendum — Multi-Omics Target Discovery in Parkinson's Disease

Krupa Patel · BIOT 6900 · Addendum added October 27, 2026 in response to the October 25 rubric-check announcement. The original report (Parkinsons Multi Omics Report.pdf) is unchanged.

A. Discordant gene: PBXIP1

Summary Data: PBXIP1 ranks #2 in the integrated list. Its RNA is decreased in PD substantia nigra ($\log_2$FC −1.59, p = 0.033), while its protein is increased ($\log_2$FC +0.20, flagged significant by Licker et al.), and it has strong genetic support ($-\log_{10}$ p = 38.7).

According to the Human Protein Atlas, PBXIP1 exhibits widespread cytoplasmic expression across both neuronal and glial cell populations (astrocytes and oligodendrocytes) in the basal ganglia. Because the transcriptomic (Lesnick) and proteomic (Licker) datasets were derived from entirely separate patient cohorts, differences in the degree of neurodegeneration and glial expansion between these cohorts offer a plausible biological explanation for the observed discordance. In post-mortem tissue, variable ratios of dopaminergic neuronal loss versus reactive gliosis across distinct clinical samples can alter bulk measurements, causing RNA and protein levels to shift in opposing directions despite accurate measurements within both individual studies. Although post-transcriptional mechanisms like translation buffering or protein stabilization could theoretically contribute, cohort-level differences in cell-type composition represent a more parsimonious explanation given the unmatched nature of the patient cohorts.

Additionally, PBXIP1's reported RNA decrease must be interpreted with caution: its unadjusted p-value of 0.033 corresponds to a Benjamini–Hochberg adjusted p-value of 0.771, and its primary probe (207838_x_at) is an Affymetrix cross-hybridizing probe, indicating that the observed transcriptomic change may simply represent platform noise rather than true biological discordance. An ideal follow-up experiment would utilize single-nucleus RNA sequencing alongside spatial proteomics on the exact same patient tissue samples to definitively resolve cell-type specific expression dynamics.

Concordance Analysis: Across all 34 genes, only 16 (47%) were concordant, and the Spearman correlation between RNA and protein $\log_2$FC was −0.136.

With a sample size of 34 genes, this 47% concordance rate and small negative Spearman correlation (ρ = −0.136) are statistically indistinguishable from zero, indicating that there is no detectable agreement between the transcriptomic and proteomic layers in these unmatched patient cohorts.

B. RNA significance after multiple-testing correction

Summary Data: No probe in GSE7621 passes multiple-testing correction: the smallest Benjamini–Hochberg adjusted p-value across all 54,318 probes is 0.538. LMNA's p = 0.0013 is unadjusted; its adjusted p-value is 0.538. Its strongest probe (244225_x_at) is an "_x_at" probe, which Affymetrix flags as potentially cross-hybridizing, and LMNA's other probes show smaller changes in mixed directions.

This absence of statistical significance after multiple-testing correction undermines LMNA's top ranking and selection as a candidate target, demonstrating that the transcriptomic layer suffers from an elevated false-positive rate. As a result, the transcriptomic layer cannot be relied upon as strongly as the genetic layer without stringent error control. However, enforcing a strict filtering threshold of adjusted $p < 0.05$ on GSE7621 would eliminate all 54,318 probes, removing the RNA layer entirely from downstream integration. To resolve this, future iterations of the discovery pipeline should either down-weight the transcriptomic layer relative to genomics or replace microarray datasets with larger, higher-powered RNA-seq cohorts capable of surviving false discovery rate correction.

C. Genomic location of PBXIP1 and LMNA

Summary Data: The GWAS signals for PBXIP1 (chr1 ~154.95 Mb) and LMNA (chr1 ~156.13 Mb) both come from the same study and lie in chromosome 1q21–q22, the region that also contains GBA1, one of the strongest PD risk genes.

Could the apparent genetic associations for PBXIP1 and LMNA represent distinct, independent risk loci, or are they secondary signals resulting from extended linkage disequilibrium tagging the nearby GBA1 locus? Performing fine-mapping and conditional statistical analyses within the 1q21–q22 region would clarify whether PBXIP1 and LMNA harbor independent causal variants or reflect locus-wide LD structure.

D. Open Targets is not independent confirmation

Citing Open Targets as independent confirmation introduces circular reasoning because Open Targets draws its genetic evidence largely from the GWAS Catalog, which served as the primary input source for this candidate list. Consequently, finding high Open Targets scores for 13 of the 15 candidate genes reflects database overlap and cross-referencing rather than orthogonal validation from an independent cohort or assay.

E. References

The Open Targets reference in the original report was cut off at the bottom of page 3, and the GWAS Catalog was not cited. Complete references:

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